

Question 1 (15 marks)**Marks**

- (a) Given the complex numbers $\alpha = 2 + 3i$ and $\beta = 5 + i$, simplify the following, leaving your answers in $x + iy$ form.
- (i) $\alpha\beta$ 1
- (ii) $\overline{\alpha + \beta}$ 1
- (iii) $\frac{1}{\overline{\alpha} - \beta}$ 2
- (b) Write the following complex numbers in modulus – argument form.
- (i) $z_1 = -9$ 1
- (ii) $z_2 = 12i$ 1
- (iii) $z_3 = (3 + 3i)^5$ 2
- (c) Solve $z^2 - (2 + i)z + (1 + i) = 0$ over the complex field. 2
- (d) A complex number z is such that $\arg(z + 2) = \frac{\pi}{6}$ and $\arg(z - 2) = \frac{2\pi}{3}$. 3
Find z , in form of $a + ib$, where a and b are real numbers.
- (e) Use De Moivre's theorem to prove that $\sin 3t = 3\cos^2 t \sin t - \sin^3 t$. 2

Question 2 (15 Marks)**START A NEW PAGE**

- (a) P_1 and P_2 are points representing the complex numbers z_1 and z_2 as shown on the Argand diagram.

If OP_1P_2 is an isosceles triangle with $\angle P_1OP_2 = 90^\circ$, show that $z_1^2 + z_2^2 = 0$.

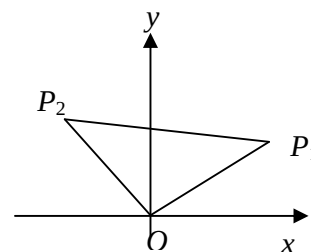


Diagram not to scale

- (b) If the point P , represents the complex number z , which lies on the unit circle about the Origin, by using the triangle inequality, or otherwise, show that: 2
 $|z^2 + z + 1| \leq 3$.
- (c) Find $\arg(z)$ if $z = \frac{-2}{1 + i\sqrt{2}}$. Give your answer correct to 4 decimal places. 3

Question 2 continues on the next page.

Question 2 continued**Marks**

- (d) $z = (1 + i)$ is one root of the equation $z^3 + pz^2 + qz + 6 = 0$, where p and q are real numbers.
- (i) Find the other 2 roots of the equation. 2
- (ii) Hence, or otherwise find the values of p and q . 2
- (e) **WRITE YOUR ANSWERS TO THIS PART OF QUESTION 2 ON THE PAGE PROVIDED AT THE END OF THE EXAM BOOKLET.**
- Given $z_1 = -3 + 2i$ and $z_2 = 1 + 4i$,
- (i) Draw neatly on the Argand diagram provided, the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ if A and B are the points representing the complex numbers z_1 and z_2 . 1
- (ii) On this Argand diagram, indicate the point C representing $z_1 - z_2$. 1
- (iii) Find the $|z_1 - z_2|$ and $\arg(z_1 - z_2)$. 2

Question 3 (15 Marks) **START A NEW PAGE**

- (a) If $|z| = 3$ and $\arg z = \theta$ determine
- (i) $\left| \frac{i}{z^2} \right|$ 1
- (ii) $\arg\left(\frac{i}{z^2}\right)$ 2
- (b) On an Argand diagram, neatly shade the region that holds simultaneously for $|z - (2 + i)| \leq \sqrt{5}$ and $\arg z < \frac{\pi}{12}$. 3
- (c) (i) Show that the solutions of $z^6 + z^3 + 1 = 0$ are contained in the solutions of $z^9 - 1 = 0$. 1
- (ii) Neatly sketch the **nine** solutions of $z^9 - 1 = 0$ on an Argand diagram, showing all important features. 3
- (iii) Mark **clearly** on your diagram the **six** roots: z_1, z_2, z_3, z_4, z_5 and z_6 of $z^6 + z^3 + 1 = 0$. 2
- (iv) Hence show that the sum of the six roots of $z^6 + z^3 + 1 = 0$ is given by $2\left(\cos\frac{2\pi}{9} + \cos\frac{4\pi}{9} - \cos\frac{\pi}{9}\right)$. 3

Question 4 (15 Marks) START A NEW PAGE**Marks**

- (a) For which values of c does $x^2 + 4x + c$ have two complex conjugate roots? **2**
- (b) If ω is one of the complex cube roots of unity, evaluate by simplifying $(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^8)$. **3**
- (c) The points A_1, A_2 and A_3 represent the complex numbers α_1, α_2 and α_3 respectively where $\alpha_1\alpha_3 = \alpha_2^2$ **3**
- Show that OA_2 bisects the angle A_1OA_3 , where O is the origin.
- (d) Triangle OAB is scalene. External equilateral triangles ABF, BOD and OAE are constructed on the sides of $\triangle OAB$. The triangles are positioned on the Argand diagram as shown.

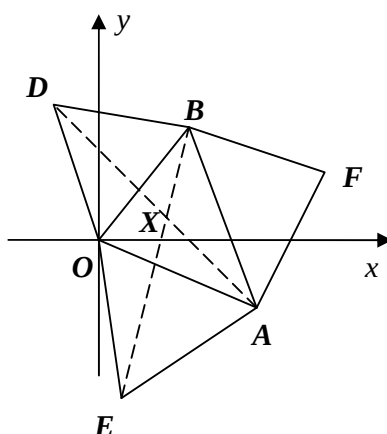


Diagram not to scale

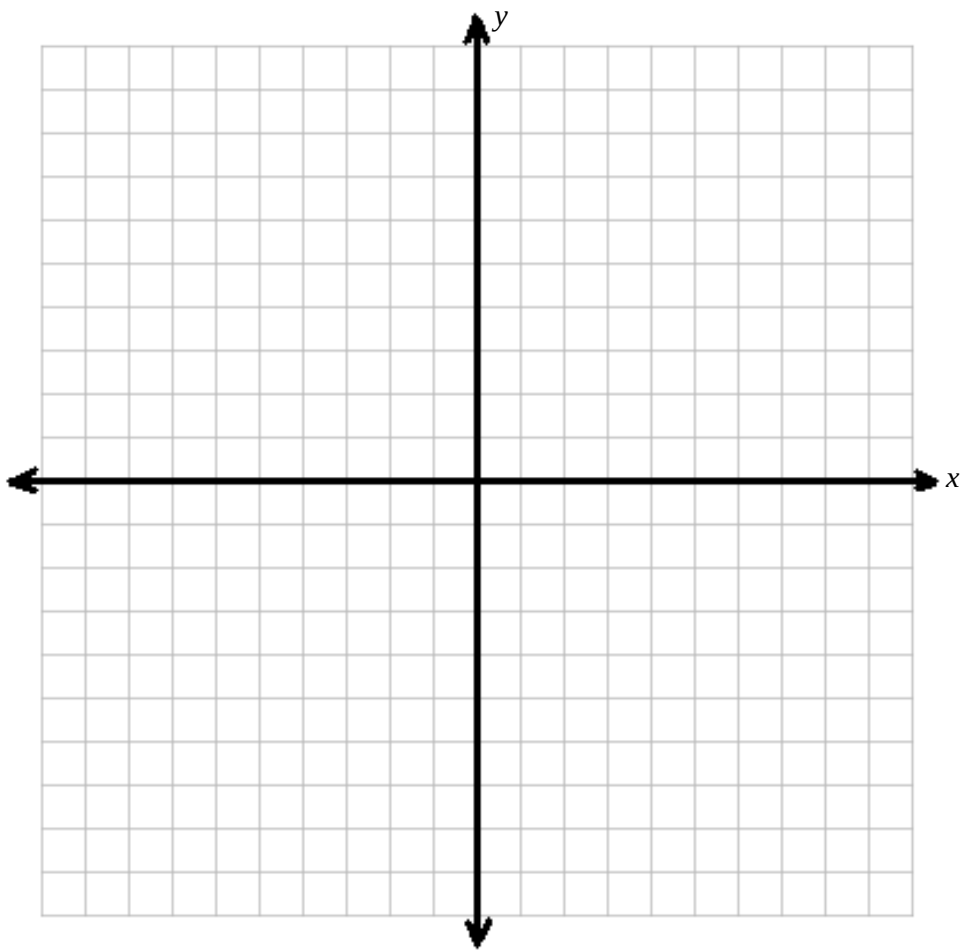
The points A, B, D and E represent the complex numbers α, β, δ and ε respectively.

Let $w = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$.

- (i) Show that $1 - w = -w^2$. **1**
- (ii) Explain why $\delta = w\beta$. **1**
- (iii) State the complex number α , represented by the point A , in terms of ε . **1**
- (iv) Hence, show that the complex number represented by the point F is $-w^2(\beta - \varepsilon)$. **2**
- (v) Hence, show that $AD = BE = OF$. **2**

Question 2 (e) ~ STAPLE TO THE BACK OF YOUR ANSWER SHEET
FOR QUESTION 2.

(i) and (ii)



(iii) _____

Y12 M.EXT2 ASSESS TASK 1
TERM 4, 2010

TERM 4 2010 MATHEMATICS Extension 2: Question ... 1 ...

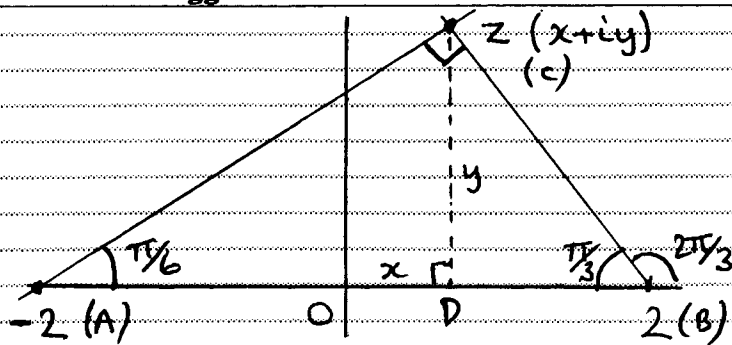
Suggested Solutions	Marks	Marker's Comments
a) i) $\alpha\beta = (2+3i)(5+i)$ $= \underline{\underline{7+17i}}$	1	
ii) $\alpha+\beta = 7+4i$ $\therefore \overline{\alpha+\beta} = \underline{\underline{7-4i}}$	1	
iii) $\frac{1}{\alpha-\beta} = \frac{1}{(2-3i)-(5+i)}$ $= \frac{-1}{3+4i}$ $= \frac{-(3-4i)}{(3+4i)(3-4i)}$ $= \underline{\underline{\frac{-(3-4i)}{25}}}$ $= \underline{\underline{\frac{-3}{25} + \frac{4}{25}i}}$	1	1/2 mark deducted if left with long vincula.
b) i) $z_1 = \underline{\underline{9(\cos \pi + i \sin \pi)}}$	1	
ii) $z_2 = \underline{\underline{12(\cos \pi/2 + i \sin \pi/2)}}$	1	1/2 mark docked (once) if cis form left in final answer.
iii) $3+3i = 3\sqrt{2}(\cos \pi/4)$ $\therefore (3+3i)^5 = (3\sqrt{2})^5 (\cos 5\pi/4) \text{ (de Moivre)}$ $= \underline{\underline{972\sqrt{2}(\cos 5\pi/4 + i \sin 5\pi/4)}}$	1	1/2 mark docked if modulus not expanded.
c) $z = \frac{(2+i) \pm \sqrt{(2+i)^2 - 4(1+i)}}{2}$ $= \frac{(2+i) \pm \sqrt{-1}}{2}$ $= \frac{(2+i) \pm i}{2}$ $= \underline{\underline{(1+i) \text{ or } 1}}$	1	

Suggested Solutions

Marks

Marker's Comments

d)



On a clear diagram, it is simple to show

$$\angle OBC = \pi/3$$

$$\angle AZB = \pi/2 \text{ (Angles of } \triangle ABC)$$

Using simple trig.

$$\frac{y}{x+2} = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}} \text{ (triangle } AZC)$$

$$\Rightarrow y = \frac{x}{\sqrt{3}} + \frac{2}{\sqrt{3}} \quad - (1)$$

$$\frac{y}{2-x} = \tan \frac{\pi}{3} = \sqrt{3} \text{ (triangle } BZC)$$

$$\Rightarrow y = 2\sqrt{3} - x\sqrt{3} \quad - (2)$$

Solving (1) and (2)

$$x=1, y=\sqrt{3} \quad \therefore \underline{z = 1+i\sqrt{3}}$$

e) $(\cos t + i \sin t)^3 = (\cos 3t + i \sin 3t)$ (de Moivre)

Expanding left hand side

$$\begin{aligned} \cos^3 t + 3i \cos^2 t \sin t - 3 \cos t \sin^2 t - i \sin^3 t \\ = \cos 3t + i \sin 3t \end{aligned}$$

Equating imaginary parts

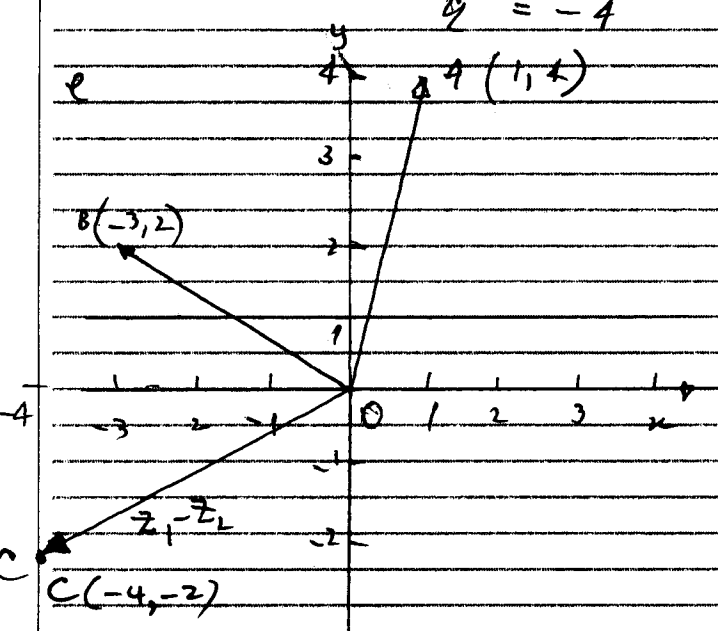
$$\underline{\underline{\sin 3t = 3 \cos^2 t \sin t - \sin^3 t}}$$

gave 1/2 for a good diagram if nothing else.

1/2 off if no mention of de Moivre

1/2 off if reason not given for plucking out right answer.

44 - 2010 - T4 MATHEMATICS Extension 2: Question.....²

2 (a) $\angle POQ = \frac{\pi}{2}$ Suggested Solutions	Marks	Marker's Comments
$ z_1 = z_2 $	$\frac{1}{2}$	
$\Rightarrow z_2 = 1z_1$	$\frac{1}{2}$	
$z_2 = -z_1$	1	
$z_1^2 + z_2^2 = 0$		
(b) $ z^2 + z + 1 \leq z^2 + z + 1 $ (triangle inequality)	1	
$\leq 1 + 1 + 1$	1	
$ z = 1$		
(c) $\text{Arg}\left(\frac{-2}{1+i\sqrt{2}}\right) = \text{Arg} -2 - \text{Arg}(1+i\sqrt{2})$	1	
$= \pi - \tan^{-1}\sqrt{2}$	1	
$= 2.1863$	1	
(i) $z_1 = 1+i$ $z_2 = 1-i$ $z_3 = -3$	1	Some students get incorrect $\angle$ since no diagram
Roots conjugate pairs real coeff. Product = -3	1	No need to rationalize
(ii) $z_1 + z_2 + z_3 = 1+i + 1-i - 3$	1	
$= -1$		
$z_1 z_2 + z_1 z_3 + z_2 z_3 = (1+i)(1-i) - 3(1+i) - 3(1-i)$	1	
$= -4$		
	1	
show direction $z_1 - z_2$		
(iii) $z_1 - z_2 = -4 - 2i$	1	
$ z_1 - z_2 = 2\sqrt{5}$		
$\text{Arg}(z_1 - z_2) = -\pi + \tan^{-1}\left(\frac{1}{2}\right) = -2.6779$	1	

2010, T4

Y11

MATHEMATICS Extension 2: Question 3

Suggested Solutions

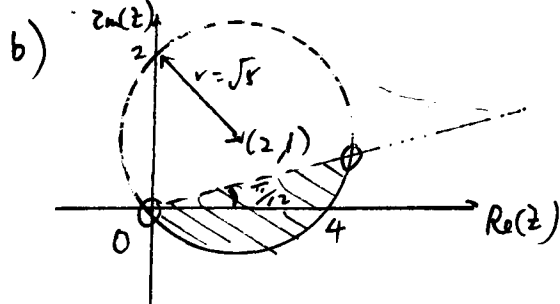
Marks

Marker's Comments

a) $\frac{1}{q}$

∴ $\arg z - 2\arg z$

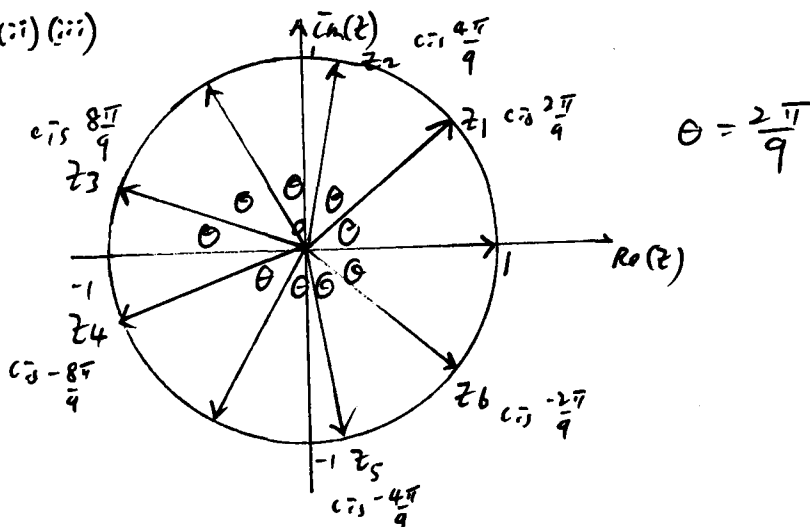
$= \frac{\pi}{2} - 2\theta$



c) $(z^3 - 1)(z^6 + z^3 + 1) = z^9 - 1 = 0$

 $z^6 + z^3 + 1$ is a factor of $z^9 - 1$ ∴ solns of $z^6 + z^3 + 1 = 0$ are contained in
solns of $z^9 - 1 = 0$

(i) (ii)



(iv) Sum of roots = $(\text{cis } \frac{2\pi}{9} + \text{cis } \frac{4\pi}{9} + \text{cis } \frac{6\pi}{9} + \text{cis } \frac{8\pi}{9} + \text{cis } \frac{10\pi}{9} + \text{cis } \frac{12\pi}{9} + \text{cis } \frac{14\pi}{9} + \text{cis } \frac{16\pi}{9}) + (-1)$

$$= 2(\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9})$$

$$= 2(\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} - \cos \frac{\pi}{9})$$

$$\text{∴ } \cos \frac{8\pi}{9} = -\cos \frac{\pi}{9}$$

1 m

1 m

1 m

3 m

no half mark
} well done

1 m circle with centre (2, 1)
radius $\sqrt{5}$, must pass
origin

1 m dotted line: $\arg z = \frac{\pi}{12}$
(15° using protractor)

1 m correct region
hole at (0, 0) included
most students forgot
dotted upper circle $-\frac{1}{2}$ m

1 m

3 m

1 m unit circle with
intercepts

1 m 9 points marked
with vectors with
 $\theta = \frac{2\pi}{9}$ at centre
of circle (many forgot
to write $\theta = \frac{2\pi}{9}$ $-\frac{1}{2}$ m)

1 m measure $\frac{2\pi}{9}$ (40°)
with protractor.
(some v. poor measurement)

2 m

6 roots labeled $z_1, z_2, z_3, z_4, z_5, z_6$
1-2 wrong root -1 m

1 m

roots in conjugate pairs

1 m

Generally z_1, z_2, z_3, z_4
are well done.

1 m

VII M. EXT 2 ASSESSMENT TASK 1
TERM 4, 2010

MATHEMATICS Extension 2: Question 4

Suggested Solutions

Marks

Marker's Comments

Q4(a) $x^2 + 4x + c$

METHOD I: For complex/unreal roots $\Delta < 0$

i.e. $4^2 - 4 \times 1 \times c < 0$

$c > 4$



METHOD II: $x^2 + 4x + c = (x+2)^2 + c - 4 = 0$

As $(x+2)^2 \geq 0 \forall x$

to be complex $x^2 + 4x + c > 0 \forall x$

$\therefore c - 4 > 0$ only

i.e. $c > 4$

METHOD III: Let α be the complex root: $\alpha = a + ib$; $a, b \in \mathbb{R}$ $b \neq 0$

$\alpha + \bar{\alpha} = 2a = -\frac{b}{a} = -4 \therefore a = -2$

$\alpha \bar{\alpha} = a^2 + b^2 = \frac{b^2}{a^2} = c$

$4 + b^2 = c \therefore b^2 = c - 4$

As $b^2 > 0 \therefore c - 4 > 0 \Rightarrow c > 4$

$\frac{1}{2}$
 $\frac{1}{2}$
1

2

(b) ω is a complex root

$\therefore \omega^3 = 1$; $\omega^2 + \omega + 1 = 0$

so $\omega^4 = \omega$; $\omega^5 = (\omega^3) \times \omega^2 = \omega^2$

so $(1-\omega)(1-\omega^2)(1-\omega^4)(1-\omega^5)$

$= (1-\omega)(1-\omega^2)(1-\omega)(1-\omega^2)$

$= [(1-\omega)(1-\omega^2)]^2$

$= [1 - \omega - \omega^2 + \omega^3]^2 = [2 - \omega - \omega^2]^2$

$= [2 - (\omega + \omega^2)]^2$

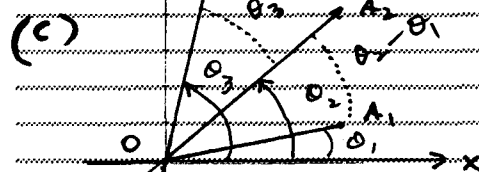
$= [2 - (-1)]^2 = 3^2$

$= 9$

$\frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$ if use both
 $\frac{1}{2}$

3



WLOG!

$\text{Arg } \alpha_1 = \theta_1$

$\alpha_1 = r_1 \text{cis } \theta_1$

$\text{Arg } \alpha_2 = \theta_2$

$\alpha_2 = r_2 \text{cis } \theta_2$

$\text{Arg } \alpha_3 = \theta_3$

$\alpha_3 = r_3 \text{cis } \theta_3$

case 1
" $\theta_3 > \theta_2 > \theta_1$ "

(2) Diagram only!

3

APPROACH I: As $\alpha_1 \alpha_3 = \alpha_2^2$

take Arg of both sides

$\text{Arg } \alpha_1 + \text{Arg } \alpha_3 = \text{Arg } \alpha_2^2 = 2 \text{Arg } \alpha_2$

i.e. $\theta_1 + \theta_3 = 2\theta_2$

i.e. $\theta_3 - \theta_2 = \theta_2 - \theta_1$

i.e. $\angle XOA_3 - \angle XOA_2 = \angle XOA_2 - \angle XOA_1$

i.e. $\angle A_3OA_2 = \angle A_2OA_1$

$\therefore OA_2$ bisects $\angle A_1OA_3$ qed.

Suggested Solutions

Marks

Marker's Comments

(c) continued

APPROACH II : see I

$$\therefore \alpha_2 = \frac{\alpha_1 + \alpha_3}{2}$$

$$\therefore \angle A_2OA_3 = \alpha_3 - \alpha_2 = \alpha_3 - \frac{1}{2}(\alpha_1 + \alpha_3) = \frac{1}{2}(\alpha_3 - \alpha_1)$$

$$\text{and } \angle A_1OA_2 = \alpha_2 - \alpha_1 = \frac{1}{2}(\alpha_1 + \alpha_3) - \alpha_1 = \frac{1}{2}(\alpha_3 - \alpha_1)$$

$$\therefore \angle A_2OA_3 = \angle A_1OA_2$$

$\therefore OA_2$ bisects $\angle A_1OA_3$

APPROACH III : As $\alpha_1\alpha_3 = \alpha_2^2$

$$\therefore \frac{\alpha_3}{\alpha_2} = \frac{\alpha_2}{\alpha_1} \quad \alpha_1, \alpha_2 \neq 0 + 0i$$

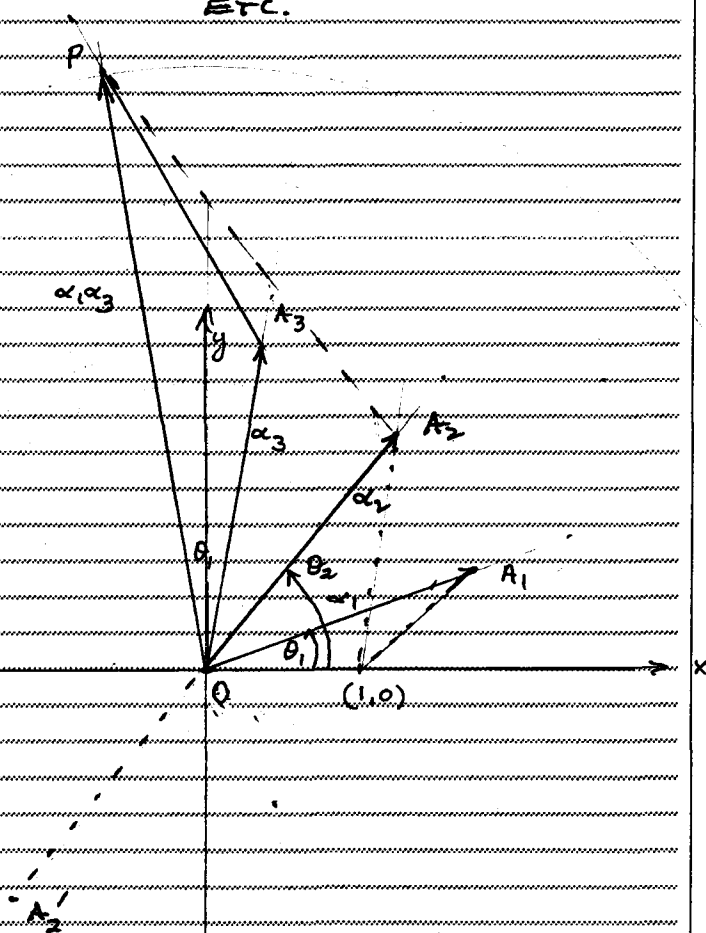
$$\therefore \arg \frac{\alpha_3}{\alpha_2} = \arg \frac{\alpha_2}{\alpha_1}$$

$$\text{i.e. } \arg \alpha_3 - \arg \alpha_2 = \arg \alpha_2 - \arg \alpha_1$$

$$\text{i.e. } \alpha_3 - \alpha_2 = \alpha_2 - \alpha_1$$

ETC.

CONSTRUCTION !!



If

$$\alpha_2^2 = \alpha_1\alpha_3$$

$$\alpha_2 = \sqrt[4]{\alpha_1\alpha_3} \text{ or } -\sqrt[4]{\alpha_1\alpha_3}$$

$$\text{Let } \arg \alpha_1\alpha_3 = \theta$$

$$\therefore \arg \alpha_2 = \frac{1}{2}\theta \text{ or } \pi - \frac{1}{2}\theta$$

by construction of III Δ s

$$\theta \equiv \alpha_3 + \alpha_1$$

$$\therefore \arg \alpha_2 = \frac{1}{2}(\alpha_3 + \alpha_1)$$

$$\text{or } \pi - \frac{1}{2}(\alpha_3 + \alpha_1)$$

$$\text{i.e. } \theta_2 = \frac{1}{2}(\theta_3 + \theta_1)$$

$$\text{or } \pi - \frac{1}{2}(\theta_3 + \theta_1)$$

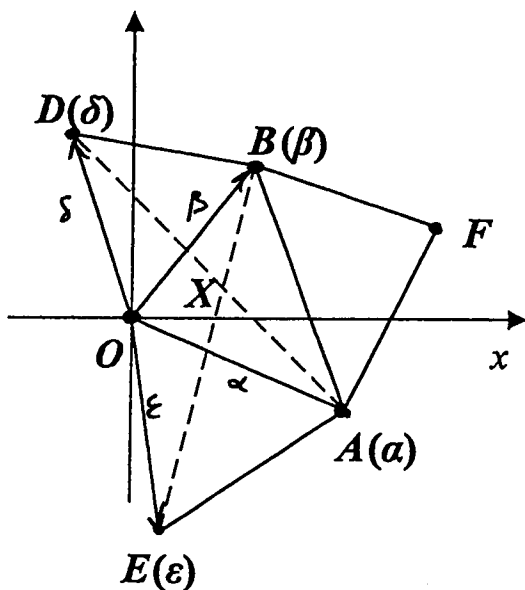
Now to show OA_2 bisects !!?

Suggested Solutions

Marks

Marker's Comments

(d)



(i) $w = \text{cis } \frac{\pi}{3}$
 $\therefore w^3 = \text{cis } \pi$ (De Moivre's Thm)

$= -1$
 ie $w^3 + 1 = 0$
 $(w+1)(w^2 - w + 1) = 0$
 as $w \neq -1$ $w^2 - w + 1 = 0$
 ie $1 - w = -w^2$ qed.

METHOD II: LHS: $1 - w = 1 - \text{cis } \frac{\pi}{3} = 1 - \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$
 $= \frac{1}{2} - i\frac{\sqrt{3}}{2}$

RHS: $-w^2 = -(\text{cis } \frac{\pi}{3})^2 = -\text{cis } \frac{2\pi}{3}$
 $= -[\cos \frac{2\pi}{3} + i\sin \frac{2\pi}{3}]$
 $= -[-\cos \frac{\pi}{3} + i\sin \frac{\pi}{3}]$
 $= \cos \frac{\pi}{3} - i\sin \frac{\pi}{3}$
 $= \frac{1}{2} - i\frac{\sqrt{3}}{2}$

$\therefore 1 - w = -w^2$ ($= \bar{w} = \frac{1}{w}$) !!!

Many ways to show (i)

$\frac{1}{2}$

(De Moivre's Thm)

ASTC

$\frac{1}{2}$

□

(ii) As $\triangle OBD$ is equilateral Δ where each angle is $\frac{\pi}{3}$ ✓ $\frac{1}{2}$
 and $OB = OD$ ie $|\beta| = |\delta|$

$\therefore \vec{OD} = \delta$ is the anticlockwise rotation by $\frac{\pi}{3}$ from $\vec{OB}$ ie β
 $\frac{1}{2}$

$\therefore \delta = \beta \text{cis } \frac{\pi}{3} = \beta w = w\beta$ qed.

□

(iii) similar argument: $\alpha = \epsilon \text{cis } \frac{\pi}{3} = w\epsilon$ ✓

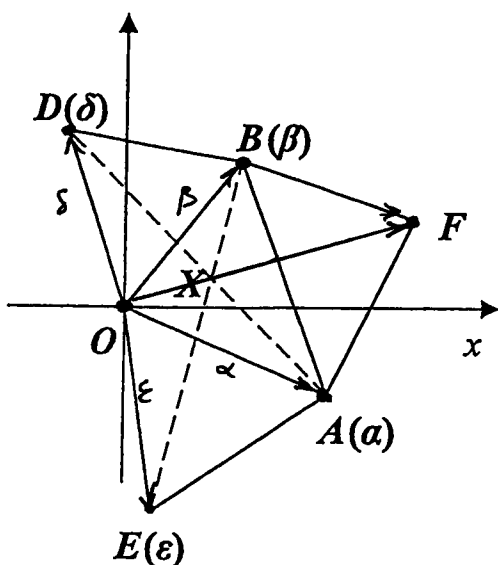
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Suggested Solutions

Marks

Marker's Comments

(d) (iv)



METHOD I: $\vec{OF} = \vec{OB} + \vec{BF}$ $\frac{1}{2}$

$$= \beta + w\beta\alpha$$

$$= \beta + w(\alpha - \beta)$$

$$= (1-w)\beta + w\alpha$$

$$= (1-w)\beta + w\cdot w\epsilon$$

$$= -w^2\beta + w^2\epsilon$$

$$= -w^2(\beta - \epsilon)$$



METHOD II: $\vec{OF} = \vec{OA} + \vec{AF}$

$$= \alpha + w\alpha\beta$$

$$= w\epsilon + w(\beta - \alpha)$$

$$= w\epsilon + w\beta - w\alpha$$

$$= w\epsilon + w\beta - w\cdot w\epsilon$$

$$= w\epsilon + w\beta - \epsilon$$

$$= w\beta + (w-1)\epsilon$$

$$= -w^2\beta + w^2\epsilon$$

$$= -w^2(\beta - \epsilon)$$

$w = \text{cis}(-\frac{\pi}{3})$

$1-w = -w^2$
 $w = -w^2$

2

(v) $\vec{EB} = \beta - \epsilon \quad \therefore EB = |\beta - \epsilon|$

$\vec{AD} = \delta - \alpha$

$$= w\beta - w\epsilon$$

$$= w(\beta - \epsilon) \quad \therefore AD = |w(\beta - \epsilon)|$$

$$= |w| \cdot |\beta - \epsilon|$$

$$= |\beta - \epsilon| \quad \text{as } |w| = 1$$

$\vec{OF} = -w^2(\beta - \epsilon) \quad \therefore OF = |-w^2| \cdot |\beta - \epsilon|$

$$= |\beta - \epsilon| \quad \text{as } |w^2| = 1$$

$\therefore BE = AD = OF = |\beta - \epsilon|$

OR Prove !!! $\triangle OEB \equiv \triangle OAD \equiv \triangle OAF$

Research: Napoleon thm; Fermat point X, ...
 - Prove OF, AD and EB are concurrent

$\frac{1}{2}$

$\frac{1}{2}$

stating that:
 $|w| = |\text{cis} \frac{\pi}{3}| = 1$

$\frac{1}{2}$

$|w^2| = |w|^2 = 1$

$|w^k| = |w|^k = 1$

2